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SYSTEMATIC REVIEW Open Access

Prostate cancer and CAR-T therapy: a systematic review

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Abstract

Importance Prostate cancer (PCa) remains a leading cause of cancer-related mortality in men globally, with castration-resistant prostate cancer (CRPC) representing a particularly challenging stage of disease. Despite therapeutic advances, including androgen receptor inhibitors (ARPIs), taxane-based chemotherapy, and radiopharmaceuticals, durable responses in metastatic CRPC (mCRPC) are limited.

Objective Immunotherapy, especially chimeric antigen receptor (CAR) T cell therapy, has revolutionized hematologic malignancies, yet its application in solid tumors like prostate cancer remains under investigation. We conducted a systematic review in accordance with PRISMA guidelines to evaluate all published and registered clinical trials (2014–2024) involving CAR-T therapy in prostate cancer.

Evidence review Databases searched included PubMed, ClinicalTrials.gov, and Cochrane. Studies were screened using stringent inclusion/exclusion criteria and analyzed for trial phase, antigen targets, dosing, safety (cytokine release syndrome [CRS], neurotoxicity, dose-limiting toxicities [DLTs]), and early efficacy outcomes. Data extraction was performed independently by two reviewers with AI-assisted adjudication. Out of 32,565 records, 27 trials met inclusion, of which 8 had published outcomes.

Findings Eight early-phase trials investigated CAR-T therapies targeting PSMA, PSCA, and other tumor-associated antigens (e.g., CD70, GD2, TGFβDN). PSMA-directed CAR-T trials showed variable safety and preliminary efficacy, with some reporting PSA reductions > 50% and radiographic partial responses. CRS (Grade 1–2) occurred in up to 100% of patients, with isolated cases of neurotoxicity and fatal adverse events in higher-dose cohorts. Trials employing armored CARs (e.g., TGFβDN, iCasp9 switches) demonstrated improved persistence and early anti-tumor activity. The use of dual-target CARs and gene-editing strategies (e.g., piggyBac transposons, PD-1 knockouts) reflect the field's evolving precision.

Conclusions and relevance CAR-T therapy in prostate cancer, particularly mCRPC, is a promising yet nascent field marked by heterogeneous safety profiles and early signs of efficacy. PSMA remains the most validated target, though antigen escape and tumor microenvironment barriers continue to limit success. Innovative constructs and combinatorial strategies may enhance clinical outcomes. Continued trial development and refinement are critical to unlocking CAR-T's potential in solid tumors.

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Key Points

Question: What are the current advancements and challenges in CAR-T cell therapies for metastatic castration-resistant prostate cancer (mCRPC)?

Findings: Multiple phase I/II trials targeting various antigens such as PSMA, PSCA, CLDN6, STEAP2, EpCAM, and others demonstrate partial responses and disease stabilization with variable safety profiles. Novel strategies include PD-1 blockade, incorporation of costimulatory domains like 4-1BB, and targeting unique T-cell subsets such as γδ T cells. Challenges remain with CAR-T cell persistence, tumor microenvironment resistance, and immune-related toxicities.

Meaning: CAR-T therapy for mCRPC shows promise but requires multimodal approaches and further optimization to overcome tumor heterogeneity and immunosuppressive microenvironments for improved efficacy and safety.

Keywords CAR-T cell therapy, Prostate cancer, PSMA, CRPC, Immunotherapy, Clinical trials, Tumor microenvironment, TGFβDN, Cytokine release syndrome

Introduction

Prostate cancer (PCa) is one of the most prevalent malignancies among men, representing a major global health concern [1]. While initial disease control is often achieved with androgen deprivation therapy (ADT), most patients progress to castration-resistant prostate cancer (CRPC) within two years, with metastatic CRPC (mCRPC) representing a particularly challenging clinical stage [2, 3]. Although newer therapies, such as abiraterone, enzalutamide, docetaxel, cabazitaxel, and radiopharmaceuticals, have extended survival, treatment resistance and toxicity continue to limit long-term efficacy [4–6].

Immunotherapy has generated excitement in oncology, but prostate cancer has proven to be particularly challenging due to its immunologically "cold" tumor microenvironment—marked by low T-cell infiltration and high expression of inhibitory receptors like Programmed Cell Death Protein 1(PD-1 [7]. While Sipuleucel-T, the only FDA-approved cancer vaccine for prostate cancer, confers modest survival benefits, immune checkpoint inhibitors such as pembrolizumab have shown limited efficacy, likely due to low Programmed Cell Death Protein Ligand1 (PD-L1) expression and immune evasion mechanisms [7, 8].

Chimeric antigen receptor (CAR) T-cell therapy, revolutionary in hematologic malignancies, is now being explored in solid tumors, including PCa [9]. This review focuses on CAR-T applications in prostate cancer, highlighting evolving strategies to enhance antigen specificity, persistence, and safety. Key targets such as prostate-specific membrane antigen (PSMA), Prostate Stem Cell Antigen (PSCA), claudin-6 (CLDN6), Six-Transmembrane Epithelial Antigen of the Prostate proteins 2 (STEAP2), CD276 (B7-H3), kallikrein 2 or prostate-specific glandular kallikrein 2 (KLK2), CD44v6, and Epithelial cell adhesion molecule (EpCAM) are under active investigation (Fig. 1) [6, 10]. Advances in CAR design, from early CD3ζ-only constructs to armored and gene-edited fifth-generation platforms, are being tailored to address the unique immune barriers of prostate tumors [11, 12].

In this systematic review, we will review emerging clinical trial data, explore novel modifications, and discuss the potential of CAR-T cell therapy to transform outcomes in treatment-refractory prostate cancer.

Materials/subjects and methods

A systematic review using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines were used, including published data in English, timeline 2014–2024 [11]. PRISMA checklist was used to ensure that all required items were addressed. The literature search databases used in this study were "ClinicalTrials.gov, "Cochrane", and "PubMed" to perform the initial extraction of all clinical trials evaluating CAR-T therapies. We used the following MeSH keywords and their combinations: Advanced PubMed term: "Chimeric antigen receptor T-cell therapy" OR "Chimeric antigen receptor therapy" OR "Chimeric antigen receptor" OR "CAR" OR "CAR T-cell therapy" OR "CAR-T" OR "CAR-T Cancer Therapy" OR "CAR-T cancer treatment" OR "CAR-T therapy". The data was last extracted up to December 1, 2024. All clinical trials with outcomes that involved Prostate cancer with CAR-T cell therapy treatment were included in the data extraction. Initial data search with keywords mentioned above retrieved a total of 27,480 from PubMed records, 3,149 from Cochrane, and 1,936 from Clinicaltrials.gov.

In the screening step, we excluded any trials that were in vitro or in vivo studies, animal studies, case reports, case series, retrospective analysis, systematic reviews, meta-analyses, and duplicate articles. We also excluded any article that lacked full text. We also excluded research studies not written in English. Two co-authors (FHZ, NS) reviewed the literature and analyzed the results independently. Rayyan AI tool served as the tiebreaker in cases

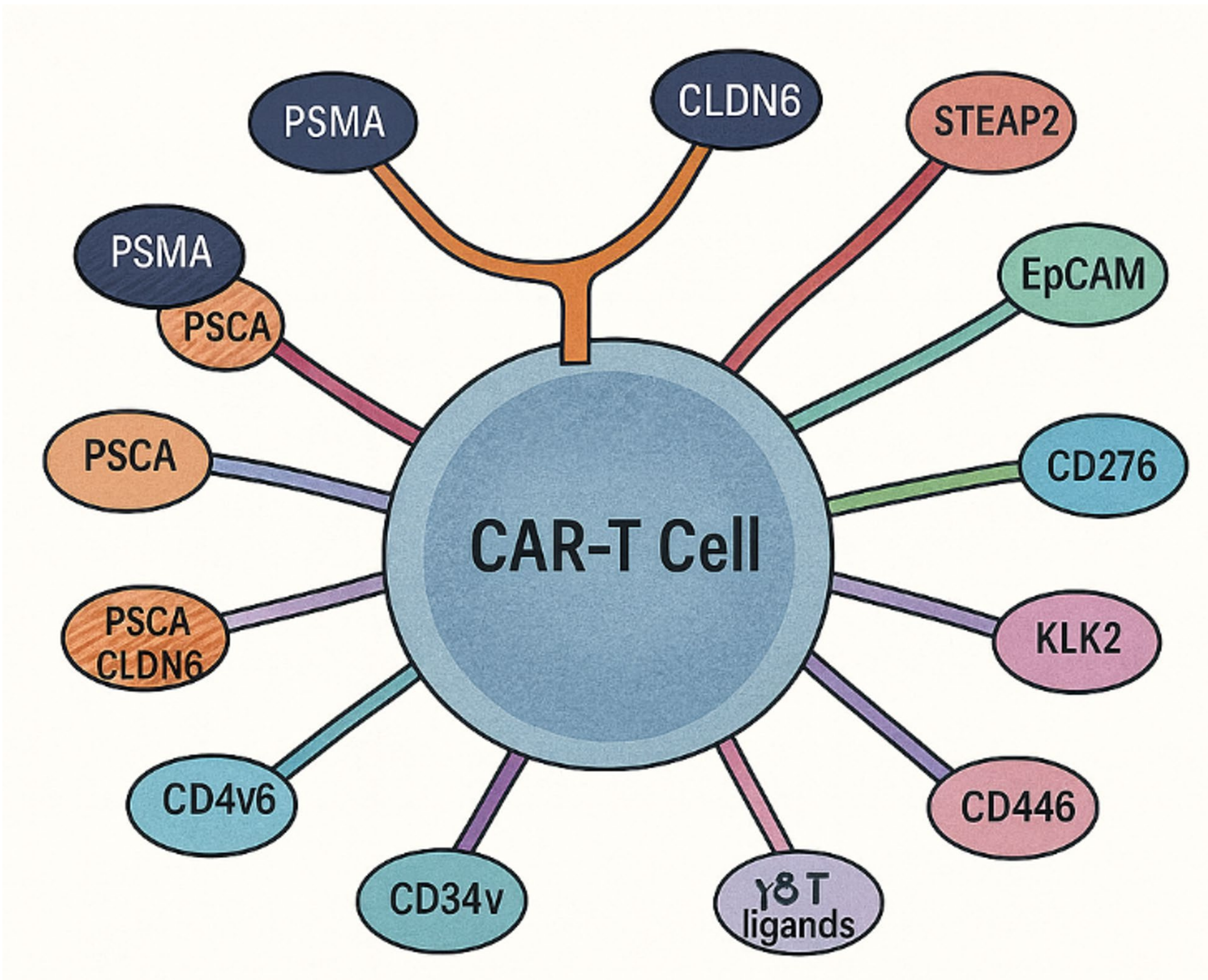

Fig. 1 Mechanism of CAR-T cell recognition of prostate cancer antigens

of disagreement in article inclusion [12]. The PRISMA flow diagram of the data collection is illustrated in Fig. 2. Of all the studies, 8 relevant CAR-T and prostate cancer clinical trials with published data were identified.

As these 8 studies were mostly in their phase I to early phase II trials, we used phase I design safety and efficacy analysis. The criteria investigated included: type of study, phase I design, number of patients, number of doses, safety evaluation periods, dose limiting toxicities (DLTs), cytokine release syndrome (CRS), and neurotoxicity. The primary endpoints of our phase I trials were focusing on overall safety (defined as complete grade 3 or higher adverse events (Grade ≥ 3 AE), neurotoxicity, and CRS) and tolerability, while preliminary efficacy result was a secondary endpoint. We also provided an overview of the trial designs, current progress, and available updates for the 19 additional ongoing trials, most of which have minimal or no published data to date.

Results

Based on our search, we have identified 5330 reports that were assessed using our inclusion and exclusion criteria. We removed duplicates and applied all the exclusion criteria using Rayyan AI tool. 27 studies were reviewed in full, 8 of which had published data from their phase 1 trial. Therefore, these results were studied in terms of endpoint safety and efficacy. Our analysis centers on the available data from current trials (Tables 1 and 2, encompassing 8 studies) and explores the future outlook of ongoing clinical trials in development (Supplemental Tables 1, 19 studies).

To delve deeper into the findings, we categorize the trials according to their target antigens, as outlined below:

CAR-T target antigens

PSMA-targeted CAR-T cells

PSMA is the most extensively investigated CAR-T target in mCRPC. Early pre-clinical studies laid demonstrated

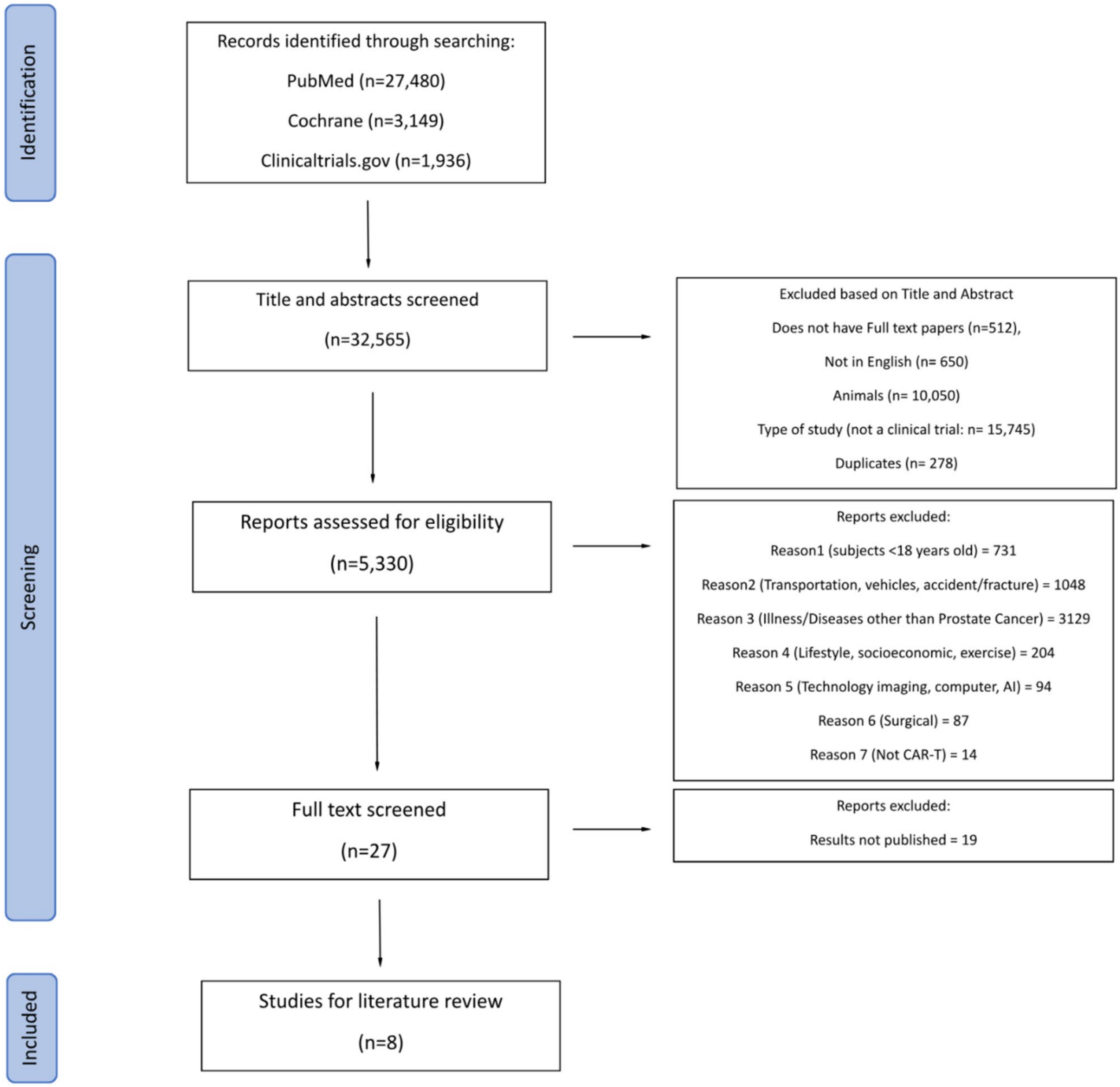

Fig. 2 Illustrating the PRISMA flow chart study selection

robust antigen-specific cytotoxicity, enhanced proliferation, and improved survival with IL-2 supplementation [22]. Early-generation PSMA CAR-T constructs demonstrated proof of concept with acceptable safety and modest efficacy.

Early-generation PSMA CAR-T trial

Initial clinical investigation in NCT00664196 trial evaluated a first-generation PSMA-targeting CAR-T given with low-dose IL-2 after Flu/Cy lymphodepletion [13, 14]. Encouragingly, in this study, no CRS, neurotoxicity, or Grade ≥ 3 AE were observed and 2/5 patients (40%) achieved partial responses (50% and 70% decline in PSA). However, the rapid loss of CAR-T after infusion was later found to be one of the contributors to the moderate reduction in PSA. This lack of persistence was subsequently attributed to the overall instability of first-generation CAR-T constructs with the state-of-the-art platforms available at that time, as well as the use of retroviral transduction rather than more advanced gene-transfer methods developed later. The limitations in finances for this trial led to its suspension. Subsequent research has focused on improving CAR design to enhance persistence and durability of response.

Table 1 Characteristics of published prostate cancer CAR-T related studies: author, year, country, study design, treatment and control group sizes, and intervention

Author, year of publication	**Slovin, S. F. (2013) [6]**	**Junghans, *R. P.* (2016) [13, 14]**	**Narayan, V. et al. (2022) [15] (Repeat 4)**	**McKean, M. et al. (2022) [16, 17]**	**Slovin, S. F. (2022) [18]**	**Mackensen, A. et al. (2023) [19]**	**Stein, M. *N.* (2024) [20]**	**Dorff, T. B. et al. (2024) [21]**
Publication location	USA	USA	USA	USA	USA	Germany	USA	USA
Registered with CT.gov	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Identifier	NCT01140373	NCT00664196	NCT03089203	NCT04227275	NCT04249947	NCT04503278	NCT02744287	NCT03873805
Trial Name	Adoptive Transfer of Autologous T Cells Targeted to Prostate Specific Membrane Antigen (PSMA) for the Treatment of Castrate Metastatic Prostate Cancer (CMPC)	Phase I Trial of Anti-PSMA Designer CAR-T Cells in Prostate Cancer: Possible Role for Interacting Interleukin 2-T Cell Pharmacodynamics as a Determinant of Clinical Response	CART-PSMA-TGFβRDN Cells for Castrate-Resistant Prostate Cancer	A Study of CART-PSMA-TGFβRDN in Patients With Metastatic Castration Resistant Prostate Cancer	P-PSMA-101 CAR-T Cells in the Treatment of Subjects With Metastatic Castration-Resistant Prostate Cancer (mCRPC) and Advanced Salivary Gland Cancers (SGC)	A Clinical Study of the Safety and Effectiveness of an Investigational Cell Therapy Given With and Without an Investigational RNA-based Vaccine in Patients With Organ Tumors	Safety and Activity Study of PSCA-Targeted CAR-T Cells (BPX-601) in Subjects With Selected Advanced Solid Tumors	PSCA-CAR T Cells in Treating Patients With PSCA+ Metastatic Castration Resistant Prostate Cancer
Phase/Status	Phase I/Not recruiting	Phase I/ Suspended (Funding)	Phase I/Not recruiting	Phase I/Terminated	Phase I/Terminated	Phase I/Recruiting	Phase I/II/ Suspended	Phase I/Recruiting
Endpoints	Primary: Safety and tolerability Secondary: efficacy, CAR-T cell persistence, changes in metastases, circulating tumor cells, biomarkers, immunity, PSA	Primary: Safety and tolerability Secondary: pharmacokinetics/pharmacodynamics of the infused drugs: dTc and IL2	Primary: Safety and feasibility Secondary: CAR T-cell distribution, biological activity, and treatment response	Primary: Safety Secondary: efficacy, feasibility, peripheral expansion and persistence of CART-PSMA-TGFβRDN	Primary: Safety, maximum tolerability dose and efficacy	Primary: Safety and tolerability, maximum tolerated dose (MTD), and recommended phase 2 dose (RP2D). Secondary: Objective response rate (ORR) and disease control rate (DCR).	Primary: Safety and tolerability. Secondary: Assessment of efficacy and characterization of the pharmacokinetics of rimiducid	Primary: safety and DLT Secondary: CAR T-cell expansion and persistence until day 28 post-infusion Exploratory: pre- and post-therapy peripheral blood immune cell subsets, tumor-infiltrating lymphocyte phenotype, and serum cytokine profile
Journal	J Clin Oncol	The Prostate	Nat Med.	J Clin Oncol	J Clin Oncol	Nat Med.	Nature Communications	Nat Med.
Impact factor/ CiteScore	45.3	4.104	58.7	45.3	45.3	58.7	14.7	58.7

Table 1 (continued)

Author, year of publication	**Slovin, S. F. (2013) [6]**	**Junghans, *R. P.* (2016) [13, 14]**	**Narayan, V. et al. (2022) [15] (Repeat 4)**	**McKean, M. et al. (2022) [16, 17]**	**Slovin, S. F. (2022) [18]**	**Mackensen, A. et al. (2023) [19]**	**Stein, M. *N.* (2024) [20]**	**Dorff, T. B. et al. (2024) [21]**
Disease	mCRPC	mCRPC	mCRPC prostate cancer	mCRPC	mCRPC	Relapsed or refractory oncofetal antigen Claudin 6 (CLDN6)-positive solid tumors including prostate cancer	castration-resistant prostate cancer (mCRPC)	Metastatic castration-resistant prostate cancer
CAR-T type	PSMA-targeted	PSMA-targeted	PSMA-targeting TGFβ-insensitive armored CAR-T cells	CART-PSMA-TGFβRDN	PSMA-targeted	CLDN6-specific	prostate stem cell antigen (PSCA)-targeted BPX-601	PSCA
CAR-T Generation	II	II	II	II	II	II	II	II/ PSCA-CAR T cells with 4-1BB co-stimulatory domain
Lymphodepletion (LD) with therapy	Yes / Cy	Yes / Flu/Cy	Yes / fludarabine/ cyclophosphamide (Flu/Cy)	Yes / Flu/Cy	Yes / Flu/Cy	Yes / fludarabine/ cyclophosphamide (Flu/Cy)	Yes/ Cy/fludarabine (Cy/Flu)	Yes / fludarabine/ cyclophosphamide (Flu/Cy)
Number of doses	2 dose levels (1 × 10^7 to 3 × 10^7 CAR + T cells/kg)	Not explicitely mentioned	4	2 patients received 1-3 × 10^7 cells 4 pts received 1-3 × 10^8 cells, 3 pts received 0.7-1 × 10^8 cells with anakinra prophylaxis	Single infusions of 0.25 ($n = 5$) to 0.75 ($n = 5$) x 10^6 cells/kg	2	9/9 (100%) patients mCRPC BPX-601 BPX-601 cell dose and Rimiducid weekly starting at day 7 at 0.4 mg/kg	3. The trial began with a starting dose level (DL1) of 100 million (M) CAR T cells without LD. Subsequent dose levels included DL2 with LD and DL3 with a reduced LD regimen.
Number of infusions	Single	Single	Single	Single and fractionated doses	Single	Single	BPX-601 cell dose once and Rimiducid weekly doses starting at day 7.	Single
Sample size	7	6 (Six patients enrolled, five treated)	13	9	10 patients as of the interim analysis. Now the trial is continued with 40 people and data is pending	22	9	14
Cohort size	7	5	18	9	40	22	9	14

Table 1 (continued)

Author, year of publication	Slovin, S. F. (2013) [6]	Junghans, *R. P.* (2016) [13, 14]	Narayan, V. et al. (2022) [15] (Repeat 4)	McKean, M. et al. (2022) [16, 17]	Slovin, S. F. (2022) [18]	Mackensen, A. et al. (2023) [19]	Stein, M. *N.* (2024) [20]	Dorff, T. B. et al. (2024) [21]
Phase design	3 + 3 dose escalation	Dose escalation	3 + 3 dose-escalation: Cohort 1: 1–3 × 10^7/m^2 CAR T-cells (no LD). Cohort 2: 1–3 × 10^8/m^2 CAR T-cells (no LD). Cohort 3: 1–3 × 10^8/m^2 CAR T-cells following cyclophosphamide (300 mg/m^2/day) and fludarabine (30 mg/m^2/day) for three days prior to infusion. Cohort – 3: 1–3 × 10^7/m^2 CAR T-cells following LD, introduced after a dose-limiting toxicity (DLT) event in Cohort 3.	3 + 3 dose-escalation	3 + 3 dose escalation planned from 0.25-15 × 10^6 cells/kg	Dose-escalation	3 + 3 dose-escalation	3 + 3 dose-escalation

Table 2 Reported efficacy and safety outcomes by target antigen: author, year, safety evaluation period, dose limiting toxicities, CRS, neurotoxicity, safety (grade ≥ 3 adverse event), stable disease, disease progression, partial or complete response, results/efficacy outcome

Author, year of publication	**Slovin, S. F. (2013) [6]**	**Junghans, *R. P.* (2016) [13, 14]**	**Narayan, V. et al. (2022) [15] (Repeat 4)**	**McKean, M. et al. (2022) [16, 17]**	**Slovin, S. F. (2022) [18]**	**Mackensen, A. et al. (2023) [19]**	**Stein, M. *N.* (2024) [20]**	**Dorff, T. B. et al. (2024) [21]**
Identifier	NCT01140373	NCT00664196	NCT03089203	NCT04227275	NCT04249947	NCT04503278	NCT02744287	NCT03873805
Safety evaluation period	4 weeks	2 years	week − 8 through end of study approximately 24 months after infusion	Up to 2 years	15 years	Median follow-up of five months (range: 29–416 days)	Through Phase 1 completion, up to 5 years	Occurrence of DLT 1–28 days Occurrence of treatment-emergent adverse events (TEAEs) up to 25 months
Dose limiting toxicities	Not reported	Not reported	1/13 (7.6%, After receiving 1–3 × 10^8/m^2 cells) Therefore dose de-escalated to 1–3 × 10^7/m^2 cells	2/9 (22.2%) had DLT at dose level of 1-3 × 10^8	1/10 (10%). macrophage activation syndrome (MAS)/ uveitis with Gr ≥ 3 CRS event	2/22 (9.1%)	2/9 (22.2%)	No DLTs were observed at DL1. At DL2, a grade 3 cystitis was noted, leading to the introduction of reduced dose DL3, where no DLTs were observed.
CRS	3/7 (42.9) Intermittent fever spikes up to 39 °C	0/5 (0%)	6/13 (46.1%)	9/9 (100%) (Grade 1–2 CRS in all patients receiving 1–3 × 10^8 cells); no CRS > Grade 2 with anakinra prophylaxis	CRS was seen in 60% (10% Gr ≥ 3) of patients.	10/22 (46%)	3/9 (33/3%)	5/14 (36%)
Neurotoxicity	Not reported	0/5 (0%)	1/13 (7.6%)	2/9 (22.2%) G5 ICANS	0/10 (0%)	1/22 (4.6%)	2/9 (22.2%)	0
Safety (Grade ≥ 3 adverse event)	Achieved in all patients	0/5 (0%)	None in Cohor 1 and 3. In Cohort 2: 3/3 (100%) of subjects received a single dose of 1–3 × 10^8/m^2 cells without LD.	Not achieved due to severe immune-mediated toxicities. 2/9 (22.2%) G5 toxicity and eventual death	The most common Grade ≥ 3 adverse events were cytopenias (60%) and infections (10%).	1/22 (4.6%)	The most frequent grade ≥ 3 TEAEs were neutropenia 5/9 (55.6%), leukopenia 4/9 (44.4%), and anemia 7/9 (77.8%). 2/9 (22.2%) treatment-related deaths occurred in the highest-dose mCRPC cohort, after which the study was terminated, without determination of the RP2D.	6/14 (43%)

Table 2 (continued)

Author, year of publication	Slovin, S. F. (2013) [6]	Junghans, *R. P.* (2016) [13, 14]	Narayan, V. et al. (2022) [15] (Repeat 4)	McKean, M. et al. (2022) [16, 17]	Slovin, S. F. (2022) [18]	Mackensen, A. et al. (2023) [19]	Stein, M. *N.* (2024) [20]	Dorff, T. B. et al. (2024) [21]
Stable disease	1/7 (14/3%) stable disease for > 6 months 1/7 (14.3%) stable scan for 16 months		5/13 (38.5%)	4/5 evaluable patients receiving ≥ 0.9×10^8 cells at day 28	not published	7/22 (31.8%)	4/6 (66.7%)	3/14 (21%)
Disease progression	4/7 (57.1%)		Not explicitly mentioned	Not explicitly mentioned	not published	8/22 (36.4%)	0/6 (0%)	1/14 (7.1%)
Partial or complete response	Partial: 2/7 (28.6%)	Partial: 2/5 (40%) with 50–70% decrease in PSA levels	3/13 (23.1%) achieved a PSA reduction of ≥ 30% 1/13 (7.6%) showing > 98% reduction in PSA	Not explicitly mentioned	Significant anti-tumor responses were observed, with declines in PSA in 7/10 (70%) patients (> 50% in 3 and > 99% in 1). Imaging showed marked to complete resolution of abnormal uptake in 3 out of 4 patients.	6/22 (27.3%) partial response. 1/22 (4.6%) complete response	Partial: 2/9 (22/2%) mCRPC patients experienced partial responses (one unconfirmed), and 56% of mCRPC patients achieved ≥ 50% reduction in PSA. 4 of which experienced a decrease in PSA levels of at least 90% (PSA90 response).	1/14 (7.1%). A partial response was observed in 1 patient, while no complete responses were reported.
Results/efficacy outcome	T cells persisted in blood for up to 2 weeks		The study demonstrated feasibility and safety of TGFβ-resistant CAR T cells, with preliminary evidence of anti-tumor activity. Future studies are needed to optimize the therapeutic approach.	Preliminary evidence of biological activity with PSA decreases in 4/7 patients, including > 50% decreases in 2/5 evaluable patients at day 28 who received ≥ 0.9×10^8 cells ($n = 7$).	Significant antitumor responses and PSA declines	The disease control rate was 67% (14 of 21), with stable disease in 7 patients. Patients with germ cell tumors treated at the higher dose level exhibited the highest response rate (ORR 57% (4 of 7)).	In mCRPC group:	PSA declines greater than 30% were observed in 4 of 14 patients. Some patients showed radiographic improvements. Limited persistence of CAR T cells was noted beyond 28 days post-infusion.

Advancement in PSMA-targeting CAR-T designs

Preclinical murine models demonstrated that using second-generation anti-PSMA with CD28 costimulation resulted in a significant decrease in tumor size [23]. CD28 is known to enhance T-cell activation and anti-tumor function [24]. Currently, a 3 + 3 dose-escalating study (NCT01140373) using this second-generation CD28-based anti-PSMA CAR-T is being conducted [6]. Seven participants received escalating doses—1×10^7 CAR-T cells/kg in cohort 1 (n = 4) and $1.5–3 \times 10^7$ CAR-T cells/kg in cohort 2 (n = 3). Treatment was generally well tolerated; all patients experienced transient fevers (three with temperatures up to 39 °C), consistent with mild cytokine release syndrome, accompanied by transient elevations in interleukin levels IL-4, IL-8, IL-6, interferon γ-induced protein 10 kDa (IP-10), and soluble interleukin-2 receptor alpha (sIL-2Rα) levels. CAR-T persistence was detected for up to two weeks post-infusion. Clinically, 1 patient achieved stable disease for over 6 months and another for 16 months, while 4 progressed. The study remains active but is no longer enrolling, with a total of 13 patients reported to date.

Innovations in gene transfer and persistence

To further enhance persistence, newer studies such as NCT04249947 evaluated P-PSMA-101 a non-viral gene transfer method with higher efficiency using a piggyBac transposon to enrich stem cell memory T cells (TSCM) and incorporate an inducible safety switch (iCasp9) [25].

Following Flu/Cy lymphodepletion, 10 patients received a single infusion across escalating doses ($0.25–15 \times 10^6$ cells/kg). The interim analysis revealed remarkable antitumor activity, with PSA declines in 7 of 10 patients (70%), including > 50% reductions in three and > 99% in one. Radiographic responses mirrored these findings, showing partial or complete regression in 3 of 4 evaluable patients, with one achieving a complete pathologic response on biopsy. Despite these encouraging results, treatment was complicated by CRS in 60% of patients (≥ Grade 3 in 10%), macrophage activation syndrome, and uveitis in one patient, and grade ≥ 3 cytopenias in 60% with infections in 10%. Notably, no neurotoxicity (ICANS) was reported. Importantly, the trial accrued its anticipated enrollment and was later terminated early not due to safety concerns, but rather, further development of this platform was not pursued for strategic and programmatic reasons. Collectively, such results underscore the potent clinical activity and translational promise of this next-generation PSMA-directed CAR-T approach.

Bi-specific PSMA CAR-T cell approach

PSMA and GD2

Few other PSMA-directed CAR-T cells clinical trials are currently ongoing for bi-specific PSMA CAR-T cell approach to mitigate antigen escape and improve specificity [26, 27]. In NCT05437315, GD2 a disialoganglioside expressed in neuroectodermal and epithelial tumors, including some prostate cancers, was targeted. The dual mechanism of targeting GD2 and PSMA may improve specificity and bypass antigen escape [28, 29].

PSMA and CD70

The approach taken for NCT05437341 focuses on Cluster of Differentiation 70 (CD70), a transmembrane glycoprotein, as a member of the tumor necrosis factor (TNF) family, associated with immune evasion and tumor progression in prostate cancer [30]. This bi-antigen targeting aims to improve recognition of heterogeneous mCRPC clones.

Microenvironment-resistant PSMA CAR-T strategies-TmPSMA-02 and TGFβDN

Given the profound immunosuppression within the prostate tumor microenvironment, recent CAR-T designs have focused on enhancing resistance to inhibitory cytokines such as transforming growth factor-beta (TGFβ), a potent driver of tumor progression, invasion, and metastasis. Several next-generation PSMA-directed CAR-T constructs have been co-engineered with dominant-negative TGFβ receptors (dnTGFBR2) to block TGFβ signaling and improve T-cell persistence and antitumor activity.

Two such ongoing trials (NCT05489991, NCT06046040) integrate dnTGFBR2 with a PD1-CD28 switch receptor and CD2 end domain to enhance T-cell activation while mitigating cytokine release and reducing the risk of CRS and macrophage activation syndrome (MAS) [31–33].

A seminal study by Narayan, V et al. (NCT03089203) evaluated TGFβ-resistant PSMA CAR-T cells in patients with metastatic castration-resistant prostate cancer (mCRPC) [15]. Among 13 treated individuals, encouraging activity was observed, with stable disease in 38.5%, PSA reductions ≥ 30% in 23.1%, and a remarkable > 98% PSA decline in one patient. However, severe CRS occurred in higher-dose cohorts, including a fatal case of grade 4 CRS and sepsis, prompting immediate protocol modification and the addition of prophylactic IL-1 blockade with anakinra. Following dose de-escalation from 1 to 3×10^8 to $1–3 \times 10^7$ cells/m^2, subsequent participants tolerated treatment without grade ≥ 3 toxicities. Despite the observed toxicities including a single death from Grade 4 CRS, this finding fell within the expected and manageable risk profile for first-in-human, dose-escalation studies of cellular therapies in advanced cancer per NCCN guideline and was appropriately addressed by prompt intervention and protocol modifications including dose de-escalation [34].

A second phase I trial by McKeen et al. (NCT04227275) also investigated CART-PSMA-TGFβRDN cells following lymphodepletion with Flu/Cy [16]. PSA reductions occurred in four of seven evaluable patients, with > 50% decline in two individuals treated at higher doses (≥ 0.9 × 10^8 cells). Although all patients experienced mild to moderate (grade 1–2) CRS, prophylactic anakinra successfully prevented severe cytokine toxicity. Nonetheless, two DLTs were observed, one grade 2 and one fatal (grade 5) event associated with multiorgan failure and MAS, resulting in early study termination. Elevated post-infusion cytokine levels, including IL-6, IL-1β, and CXCL9, were consistent with hyperinflammatory responses. Despite these setbacks, findings from both trials underscore the therapeutic potential of TGFβ-resistant PSMA CAR-T cells and highlight the critical role of optimized dosing, safety-switch mechanisms, and cytokine modulation in refining future prostate CAR-T strategies. NCT03089203 investigating CAR-T-PSMA-TGFβRDN cells remains open [15]. The NCT04227275 has informed future strategies to mitigate CAR-T-associated toxicities.

Status of current PSMA CAR-T pipeline

Multiple ongoing PSMA-targeting trials (NCT05354375, NCT04768608, NCT06228404, NCT04053062, NCT04633148, NCT04429451) are investigating CAR-T safety and efficacy in CRPC [35–40]. All of these studies are focussed on mCRPC, investigating the safety, efficacy and AE of Anti-PSMA CAR-T within a safety evaluation period ranging from 28 days to up to 15 years. NCT04768608 is exploring combined PSMA and PD-1 blockade [36]. NCT04053062 using co-expression of tumor necrosis factor superfamily member 14 (TNFSF14, an immune cell accumulation and activation facilitator within tumors) Cytokine with CAR-T cells was suspended due to lack of efficacy. Some terminated trials, NCT04633148 were investigating innovative approaches such as universal CAR-T (Inert CAR-T that only are activated with target module switch) and TM peptide PSMA were unfortunately halted due to having very limited biological activity [38, 39].

PSCA

Beyond PSMA, PSCA is highly expressed in advanced prostate cancer and other solid tumors, making it a promising target for CAR-T therapy.

Early generation PSCA CAR-T trials

Key clinical efforts evaluating PSCA-targeted CAR-T therapy in metastatic castration-resistant prostate cancer (mCRPC) include the BPX-601 trial (NCT02744287), a Phase I/II open-label study assessing a rimiducid-inducible PSCA-directed bispecific CAR-T construct [20]. This trial enrolled 24 patients with metastatic pancreatic adenocarcinoma and nine with mCRPC, who received Cy/Flu lymphodepletion followed by BPX-601 infusion (5 × 10^6 cells/kg) with or without rimiducid at varying schedules. The therapy achieved disease stabilization in two-thirds of participants and partial responses in approximately 22%, with ≥ 50% PSA declines in more than half of evaluable patients, including several achieving ≥ 90% reductions, demonstrating meaningful antitumor activity in this heavily pretreated population.

However, treatment-related toxicities limited further development. DLT occurred in 22% of participants, with CRS in 33%, neurotoxicity in 22%, and grade ≥ 3 cytopenias—neutropenia (56%), leukopenia (44%), and anemia (78%), frequently observed. Two treatment-related deaths in the highest-dose cohort led to early trial termination. Detailed case analyses revealed that both fatal events were preceded by sequential episodes of CRS, carHLH, and disseminated intravascular coagulation, complicated by infectious sepsis despite initial CRS resolution. These findings highlight the need for careful cytokine modulation and infection monitoring in future PSCA CAR-T trials.

Another active trial, NCT05805371, is investigating PSCA-directed CAR-T cells administered biweekly (50 million cells per dose, up to six doses) with or without metastasis-directed radiotherapy in mCRPC [41]. This trial aims to evaluate safety, PSA responses, CAR-T persistence, and immune correlates, with long-term follow-up extending up to fifteen years. The addition of radiotherapy may enhance CAR-T infiltration and antigen expression, potentially improving efficacy while maintaining safety in this challenging disease setting.

Bi-specific PSCA CAR-T cell therapies

PSCA and tumor necrosis factor receptor superfamily member 9 (TNFRSF9)

A newer phase I open-label trial (NCT03873805) by *Dorff, T.B. et al.* is evaluating second-generation PSCA-CAR-T cells with a TNFRSF9 (CD137/4-1BB) costimulatory domain in PSCA-positive mCRPC [21, 42]. The study primarily assesses safety and dose-limiting toxicities (DLTs), with secondary and exploratory analyses examining CAR-T expansion, persistence through day 28, and associated immune and tumor-infiltrating lymphocyte (TIL) profiles.

Three dose levels were investigated: 100 million CAR-T cells without lymphodepletion (DL1, $n = 3$), with standard Flu/Cy lymphodepletion (DL2, $n = 6$), and with reduced lymphodepletion (DL3, $n = 5$). No DLTs occurred at DL1 or DL3, while one grade 3 cystitis was observed at

DL2. CRS developed in 5/14 (36%) of patients, though all cases were manageable, and no neurotoxicity was reported. Overall, 6/14 (43%) experienced grade ≥ 3 AEs. Preliminary efficacy signals included stable disease in 21% of patients, a partial response in 7%, and PSA declines greater than 30% in 29%, with some radiographic improvements. CAR-T persistence was limited beyond 28 days. The study remains active but is no longer recruiting participants.

PSMA/PCA-targeted trials

An ongoing trial (NCT05732948) is investigating PD-1 silencing to enhance CAR-T cell activity and persistence in the tumor microenvironment [43]. Since the PD-1/PD-L1 axis is a major immune evasion pathway, this strategy may improve therapeutic outcomes [44]. Currently, the status of this dose escalation trial is unknown. The primary outcome is tumor response and the secondary outcome is AE.

Other emerging targets

A variety of other antigens are also under exploration to expand CAR-T applicability in prostate cancer.

CLDN6

Claudin 6 (CLDN6), a tight junction protein and oncofetal antigen selectively expressed in numerous solid tumors, is a promising target in prostate cancer. A nanoparticulate RNA-lipoplex (RNA-LPX) vaccine was developed to enhance CAR antigen delivery to lymphoid tissues, boosting selective CAR-T cell expansion in mouse models [45]. The ongoing trial NCT04503278 evaluates CLDN6 CAR-T cells with or without RNA-LPX in patients with relapsed/refractory CLDN6-positive solid tumors, including prostate cancer [19]. CLDN6 CAR-Ts have shown particularly promising activity, achieving a disease control rate of 67% and partial or complete responses in approximately one-third of patients, with mostly mild to moderate CRS.

Preliminary results are encouraging: following Flu/Cy lymphodepletion, treated patients experienced manageable toxicities, including cytokine release syndrome (CRS) in 46% and isolated neurotoxicity or grade ≥ 3 adverse events in only 4.6% (1/22). Among 22 evaluable patients, disease control was achieved in 67% (14/21), with partial responses in 27%, complete response in 4.6%, and stable disease in 32%. The highest overall response rate (57%) was observed in germ cell tumor cohorts at higher dose levels. These findings highlight CLDN6 as a safe, immunogenic, and clinically active target, with the trial remaining open to further enrollment and dose optimization.

STEAP2 trials

Two trials, NCT06572956 and NCT06267729, target the STEAP2 transmembrane protein in prostate cancer due to its high, uniform expression across all disease stages and limited presence in normal tissues [46, 47]. STEAP2 plays a role in tumor proliferation, migration, and invasion, making it an ideal CAR-T target with reduced risk of off-tumor toxicity. Although no results are available yet, both trials primarily assess the safety and efficacy of STEAP2-directed CAR-T in mCRPC patients.

EpCam (CD326)

Epithelial Cell Adhesion Molecule (EpCam) is a glycoprotein highly expressed in epithelial carcinoma, prostate cancer, and their metastasis [48]. Deng et al. developed an EpCAM-specific CAR-T, demonstrating tumor reduction and improved survival in PC3 prostate cancer xenograft models [49]. The NCT03013712 trial is evaluating the safety and efficacy of EpCAM CAR-T in EpCAM-positive prostate cancer, with primary outcomes focused on toxicity and adverse events, and secondary outcomes assessing survival and anti-tumor efficacy [50]. The trial has enrolled 60 participants in a dose-escalation design; current trial status is unknown.

CD276 trials

CD276 (B7-H3) is an immune checkpoint molecule that is overexpressed in various tumor tissues but minimally expressed in normal tissues [51]. It modulates immune responses and promotes tumor invasion, metastasis, metabolism, and therapy resistance, with its expression linked to poor prognosis. The NCT04691713 trial is assessing the safety, tolerability, and efficacy of CD276-targeted CAR-T cells, with overall response rate (ORR) as the primary endpoint. Currently, five patients with CD276 + solid tumors are enrolled [52].

KLK2 trials

KLK2, a serine protease overexpressed in prostate cancer, plays a key role in disease progression [53]. Phase I trial NCT05022849 investigates KLK2-targeted CAR-T cells in mCRPC patients, assessing adverse events (AE), dose-limiting toxicities (DLT), and overall response rate (ORR) [54]. This dose-escalation study includes 15 participants with up to 15 years of AE follow-up. No results are currently available.

CD44v6

CD44v6, an alternatively spliced isoform of the CD44 transmembrane glycoprotein, plays a key role in cell-cell and cell-matrix interactions and is linked to cancer stem cells, tumor progression, metastasis, and poor prognosis [55]. The NCT04427449 trial investigates the CD44v6 targeted CAR-T cells which are aimed to inactivate the signaling pathways such as Mitogen-Activated Protein Kinase, and Extracellular Signal-Regulated Kinase (MAPK/ERK) and phosphoinositide 3-kinase/protein kinase B (PI3K/Akt), which are crucial for tumor progression [56]. This dose-escalation study, with 100 participants enrolled, assesses AE, infusion safety, and clinical response. The current trial status is unknown.

γδ T cells

γδ T cells can eliminate tumor cells independently of major histocompatibility complex (MHC) presentation, allowing recognition of a wide range of tumor-associated antigens and stress-induced ligands [57]. Bridging innate and adaptive immunity, they uniquely detect ligands expressed on transformed cells [58]. Natural killer group 2 member D (NKG2D), an activating receptor on both NK and γδ T cells, enhances this anti-tumor activity [59]. CAR-engineered γδ T cells targeting NKG2D ligands (NKG2DL) exhibit enhanced cytotoxicity and specificity toward tumor cells. Preclinical models demonstrated promising anti-prostate cancer responses using NKG2DL-targeting CAR γδ T cells [60]. The ongoing phase I trial NCT04107142 is evaluating their safety and efficacy in solid tumors, including prostate cancer. Phase I/II trials continue to investigate various CAR-T strategies to overcome therapeutic challenges in mCRPC [61].

Key safety and AEs observations by targets

PSMA CAR-T

CRS occurred in up to 60% of patients in advanced trials (≥ Grade 3 in 10%), ICANS was rare, MAS was reported, and Grade ≥ 3 cytopenias and infections were common in high-dose cohorts. Microenvironment-resistant designs (TGFβRDN, PD1-CD28 switch) mitigated persistence challenges but occasionally triggered Grade 4 CRS and rare fatal events, prompting dose adjustments and prophylaxis (e.g., anakinra).

PSCA CAR-T

PSCA-targeted CAR-T cells exhibit similar toxicity profiles, with CRS rates around one-third and low-grade neurotoxicity in 20–25% of cases. In trials with TNFRSF9 costimulation, CRS occurred in ~ 36%, no neurotoxicity, and Grade ≥ 3 AEs in 43%. Bi-specific PSCA/PD-1 silencing and PSCA + radiation strategies are ongoing, with AE monitoring up to 15 years. Severe toxicity (treatment-related deaths) occurred in early BPX-601 studies due to overlapping CRS, ICANS, and sepsis.

Other emerging targets

(CLDN6, STEAP2, EpCam, KLK2, CD44v6, γδ T cells)

Initial reports show CRS in 4.6–46%, neurotoxicity < 5%, Grade ≥ 3 AEs in 4.6–43%. No consistent patterns of severe ICANS reported yet. Safety largely manageable with careful monitoring and dose escalation. CLDN6-targeted CAR-T therapies demonstrate a favorable safety profile with CRS in fewer than half of patients, mostly grade 1–2, and minimal neurotoxicity. For emerging antigens such as STEAP2, EpCAM, CD276, and KLK2, published safety data are limited, but early-phase findings suggest generally tolerable toxicity consistent with established CAR-T experiences in solid tumors. Overall, while the toxicity profile of prostate-directed CAR-T therapy remains manageable, it requires vigilant monitoring and refined cytokine control to balance efficacy and safety.

Overall efficacy and durability of response

CAR-T cell therapy for prostate cancer has demonstrated measurable antitumor activity across multiple targets, though durability of response remains a primary limitation.

PSMA-targeted CAR-T

Across reported PSMA-targeted CAR-T cell trials in mCRPC, objective antitumor activity has been observed but remains variable and often transient. Early-generation PSMA CAR-T constructs (e.g., *NCT00664196*) demonstrated isolated radiographic partial responses and PSA reductions of 50–70% in a subset of patients, but limited persistence curtailed durable benefit.

Subsequent second-generation designs incorporating CD28 costimulation (*NCT01140373*) achieved disease stabilization in several patients for up to 16 months, though PSA declines were modest. Later trials employing enhanced cellular engineering have reported higher activity: the non-viral piggyBac PSMA CAR-T (*NCT04249947*) produced PSA declines in 70% of treated patients, including ≥ 50% declines in three and > 99% in one, with radiographic partial or complete responses documented in several cases. Constructs incorporating TGF-β–resistant domains (*NCT03089203*, *NCT04227275*) yielded PSA reductions in 23–40% of evaluable patients, with occasional deep responses exceeding 90–98% PSA decline, though the majority achieved only stable disease. Despite evidence of potent on-target activity in some individuals, durability of response remains limited across studies, with CAR-T expansion and persistence frequently waning within weeks. Newer platforms designed to enrich stem-like memory (TSCM) populations and integrate safety switches show early signals of improved persistence and deeper responses, but long-term disease control remains to be conclusively demonstrated.

PSCA-targeted CAR-T

Clinical evaluation of PSCA-targeted CAR-T cell therapies in metastatic castration-resistant prostate cancer (mCRPC) remains early but has shown encouraging antitumor signals in select patients. The first-in-human phase I trial (*NCT03873805*) of BPX-601, a PSCA-directed CAR-T incorporating an inducible MyD88/CD40 costimulatory domain (iMC), demonstrated PSA declines in ~40% of evaluable participants and radiographic disease stabilization in several others, particularly when combined with rimiducid to activate the costimulatory switch. Deep PSA reductions exceeding 50% were documented in isolated cases, though overall response rates were modest. Other early-phase PSCA-directed programs (e.g., *NCT02744287*, *NCT05489954*) remain in preliminary stages, with limited data but similar trends of transient disease stabilization and occasional PSA responses. Across trials, CAR-T cell persistence has been short-lived, typically peaking within one to two weeks after infusion and declining thereafter, paralleling the limited durability of clinical responses. Combination strategies that augment T-cell costimulation or pair PSCA-CAR-T with checkpoint blockade are being explored to enhance expansion and maintain durable activity. Collectively, PSCA-targeted CAR-T therapies exhibit measurable biologic and clinical activity, yet sustained disease control remains elusive, emphasizing the need for constructs that improve persistence and microenvironmental resistance.

Other targeted CAR-Ts

Among emerging prostate cancer CAR-T targets, early clinical data suggest variable but encouraging antitumor activity, with most constructs demonstrating transient disease control rather than durable remission.

CLDN6-directed CAR-T cells (*NCT04503278*) showed the most mature results to date, with a disease control rate of 67% (14/21), including partial responses in 27% and one complete response, alongside stable disease in nearly one-third of patients. While these outcomes indicate clear clinical activity, responses were primarily short-lived, with durability data still limited by brief follow-up.

STEAP2-, EpCAM-, CD276-, KLK2-, and CD44v6-targeted CAR-T trials remain in early dose-escalation phases with no published efficacy results; preclinical and early reports nonetheless suggest potential for antigen-specific tumor regression and PSA stabilization, pending clinical confirmation.

γδ T cell–based CAR approaches (*NCT04107142*) have demonstrated robust preclinical antitumor responses and are now under early-phase clinical evaluation. These cells, capable of MHC-independent recognition of tumor antigens such as NKG2D ligands, offer the promise of broader and more durable activity, though clinical data remain preliminary.

Discussion

The treatment landscape for metastatic CRPC has broadened with multiple agents demonstrating increased overall survival benefits across specific disease states and biomarker-defined subsets. ARPIs such as abiraterone, enzalutamide, apalutamide, and darolutamide, form the backbone of systemic therapy and have majorly improved outcomes in both metastatic hormone-sensitive as well as castration-resistant subgroups [62]. Taxane-based chemotherapy remains central, with docetaxel-based chemotherapy, commonly used in patients with aggressive disease or high tumor burden and cabazitaxel providing a proven survival advantage following docetaxel progression [63].

Radiopharmaceuticals have further enhanced and diversified treatment options. For instance, radium-223 improves survival in symptomatic patients with bone-predominant disease without visceral metastases, while lutetium-177-PSMA-617 (Pluvicto) has robustly demonstrated increased survival and progression-free survival benefits in PSMA-positive mCRPC patients specifically after ARPI and taxane therapy. Furthermore, this approach has emerging data supporting activity even in taxane-naive patients. In parallel, molecular selected therapies including Poly (ADP-ribose) polymerase (PARP) inhibitors (e.g. olaparib, rucaparib, niraparib and talazoparib) have become standard for patients with homologous recombination repair (HRR) gene alterations, particularly BRCA 1/2, either as monotherapy or in combination with ARPIs.

Despite these advances, resistance, toxicity, and the limited efficacy of current immunotherapies, particularly in heavily pretreated or advanced disease, remain key challenges. Considering this unmet need, CAR-T cell therapies for mCRPC remain investigational and are not part of standard treatment algorithms yet. As demonstrated earlier, while the early phase clinical trials targeting various antigens have demonstrated promising feasibility and preliminary antitumor activity, challenges including limited T-cell persistence, immunosuppressive tumor microenvironment, antigen heterogeneity, as well as on-target/off-tumor toxicity have constrained broader applicability thus far. As such, CAR-T therapies are currently best positioned within clinical trials for select, heavily pretreated patients who have exhausted multiple approved systemic options within the standard of care. Ongoing, advances in CAR engineering, armoring strategies, and comprehensive approaches may ultimately define a unique future role for CAR-T therapy as a later-line or biomarker-enriched strategy within the evolving mCRPC therapeutic landscape.

CAR-T in prostate cancer

CAR T cell therapy, a major breakthrough in hematologic malignancies, is now being explored in prostate cancer [64]. With ongoing refinements from first- to fifth-generation CARs, including enhanced costimulatory domains, cytokine integration, and gene editing—efforts aim to overcome barriers in solid tumors [65]. One of the current challenges in prostate cancer treatment lies in the tumor's intratumoral heterogeneity, complex epigenetic landscape, and the difficulty in regulating CAR-T therapy–associated cytotoxicity [66]. The above-mentioned phase I/II CAR-T trials focus on various strategies and targets in mCRPC patients in an effort to minimize these issues. PSMA, highly expressed in advanced prostate cancer, remains the most extensively studied CAR-T target, with trials investigating both single and multi-antigen strategies to address immune evasion and tumor heterogeneity [67].

Overcoming tumor microenvironment challenges

One of the other known challenges of CAR-T therapy is persistence in tumor microenvironment. Once a CAR-T cell reaches the tumor vicinity, it needs to overcome the harsh tumor microenvironment that are known to dampen the T-cell immune responses at various levels and cause tumor resistance using antigen escape mechanisms [68]. Scientists have come up with several strategies to enhance CAR-T survival and proliferation in these harsh tumor microenvironments.

TGFβ plays a key role in T-cell suppression and inhibition of antigen presentation, facilitating tumor immune escape [69]. In a prostate cancer mouse model, combining PSMA-CAR-T cells with a dominant-negative TGFβ receptor lacking the intracellular domain (TGFβDN) reduced these suppressive mechanisms, resulting in enhanced CAR-T proliferation and persistence within the tumor microenvironment in vivo [70, 71]. These findings led to the Phase I trials NCT03089203 and NCT04227275, which evaluated PSMA-targeting, TGFβ-insensitive armored CAR-T cells in patients with mCRPC. 15,17 Trial analyses demonstrated that greater CAR-T expansion correlated with higher cytokine release and toxicity, indicating a link between immune activation and adverse events. Although NCT04227275 reported significant immune-mediated toxicities, reductions in PSA levels were observed, suggesting potential antitumor activity. Thus, PSMA-CAR-T cells incorporating TGFβDN remain a promising strategy, warranting further safety and feasibility studies.

PD-1 is another immunosuppressive pathway exploited by prostate cancer through PD-L1 expression on tumor cells. Pembrolizumab has shown some efficacy in prostate cancer by inhibiting PD-1 [72]. Building on this rationale, the completed NCT04768608 trial investigated PD-1 blockade in combination with PSMA-targeted CAR-T therapy, though results have not yet been published.

Beyond checkpoint inhibition, CAR-T design has focused on costimulatory domains such as 4-1BB (CD137/TNFRSF9) [42, 73, 74]. Preclinical studies demonstrated improved CAR-T persistence with 4-1BB–containing constructs, leading to the NCT03873805 trial targeting PSCA-positive mCRPC [21, 42]. Early results showed partial responses with > 30% PSA decline in 4 of 14 patients, though further evaluation is needed. Notably, some preclinical data suggest that substituting 4-1BB with CD2 may reduce cytokine production while preserving antitumor efficacy.

The NCT04107142 trial explored CAR-T therapy using γδ T cells, a less abundant T-cell subset comprising approximately 1–5% of peripheral blood T cells [75]. Unlike αβ T cells, γδ T cells act independently of MHC, enabling broader tumor recognition [57]. This trial aims to activate both NK and γδ T cells to target solid tumors, including prostate cancer. While preclinical findings are encouraging, clinical safety and efficacy are still under investigation [60].

Prostate stem cell antigen (PSCA) is another promising target, expressed in over 80% of prostate tumors and associated with higher Gleason scores and metastatic disease. PSCA expression is also observed in pancreatic, gastric, and bladder cancers. Preclinical studies have shown that PSCA-specific CAR-T cells effectively mediate tumor cytotoxicity and delay disease progression.

Additional antigens under investigation include claudin-6 (CLDN6), STEAP2, CD276 (B7-H3), KLK2, and CD44v6. These targets vary in specificity and expression profiles, offering diverse therapeutic opportunities. For instance, EpCAM (CD326), which is overexpressed in several epithelial malignancies including prostate cancer, has demonstrated tumor inhibition and prolonged survival in preclinical prostate cancer models using EpCAM-directed CAR-T cells, supporting its continued clinical development [65].

Despite the complexities of the solid tumor microenvironment, including physical barriers, immune suppression, and antigen heterogeneity, ongoing clinical trials are actively evaluating these targets and novel CAR-T constructs.

Safety and adverse events

Across CAR-T trials in mCRPC, the incidence and severity of AE correlate with CAR generation, dose, and microenvironment-resistance features. The most consistent toxicities observed with prostate cancer–directed CAR-T therapies are CRS, cytopenias, and, less commonly, MAS or neurotoxicity. Early-generation CARs were found to be generally well-tolerated, with minimal CRS or neurotoxicity. However, these trials demonstrated limited persistence and efficacy. Second-generation CARs incorporating costimulatory domains (CD28, 4-1BB/TNFRSF9) had improved activation and antitumor function but increased AE profile including CRS and cytopenias, prompting the implementation of prophylactic interleukins blockade (such as using anakinra) and dose de-escalation strategies. ICANS is relatively rare across prostate CAR-T studies, with most trials reporting few or no neurologic events.

Overall efficacy and durability of response

Based on the above trials, CAR-T therapy in mCRPC has progressed from early feasibility to biologically active clinical constructs with measurable efficacy. Yet, achieving durable remissions remains challenging due to poor CAR-T persistence, tumor heterogeneity, and the immunosuppressive microenvironment. Future strategies integrating dual-antigen targeting, cytokine modulation, gene editing, and safety switches are likely to further optimize the balance between efficacy, persistence, and tolerability in this evolving therapeutic landscape.

Conclusions and future direction

Prostate cancer–directed immunotherapies, including CAR-T, have made substantial strides, demonstrating that antigen-specific immunetargeting can be feasibly applied to solid tumors. Among antibody-based and adoptive approaches, CAR-T cells, along with bispecific T-cell engagers (BiTEs), and antibody-drug conjugates (ADCs) are distinct strategies with differing mechanisms, efficacy signals, and toxicity profiles in mCRPC.

Comparing clinical activity and feasibility

Across CAR-T therapy trials, PSMA- and PSCA-targeted CAR-T cells have shown the greatest proof-of-concept clinical activity, with measurable PSA declines, radiographic responses, and disease stabilization in subsets of patients. However, responses have generally been transient, with limited CAR-T persistence and sometimes substantial toxicity currently constraining durability and scalability. In contrast, BiTEs, particularly in regard to PSMA-targeted constructs have demonstrated higher and more reproduceable PSA response rates in heavily pretreated patients, albeit with near-universal CRS. Moreover, BiTEs with more off-the-shelf availability, lack of lymphodepletion, and outpatient administration are more operationally feasible compared to CAR-T, despite the need for continuous dosing and the challenges of immunogenicity.

ADCs add a third player in the paradigm, delivering cytotoxic loads directly to antigen-expressing cells without T-cell involvement. Hence, while PSA response rates with ADCs have been more modest, disease control rates are encouraging, and toxicity profiles are predictable and largely non-immune mediated, making ADCs attractive for broader clinical use.

Toxicity as a differentiating factor

While other emerging targets in CAR-T therapy such as CLDN6, STEAP2, EpCAM, KLK2, CD44v6, and γδ T cells further expand the therapeutic landscape, offering opportunities to target heterogeneous tumor populations and potentially mitigating on-target, off-tumor toxicity, current CAR-T trials need to further reduce their reported grade 4 CRS, ICANS, MAS, even in next-generation or microenvironment-resistant constructs. Comparatively, BiTEs consistently induce CRS, typically early in treatment, but severe neurologic toxicity is less frequent and often manageable with step-up dosing and cytokine-directed prophylaxis. ADC toxicity profiles are majorly driven by payload effects, most commonly neutropenia and neuropathy, and completely avoid immune-mediated syndromes. Nonetheless, dose optimization remains critical to minimize infectious complications.

Mechanisms of resistance and durability

All three technologies face overlapping resistance mechanisms, including antigen heterogeneity, antigen loss, and the profoundly immunosuppressive tumor microenvironment characteristic of mCRPC. CAR-T approaches uniquely allow genetic engineering to address such barriers. While early CAR-T trials, including first-generation PSMA, universal CAR-Ts and TM-peptide PSMA CAR-Ts suffered from limited persistence, resulting in transient responses and suboptimal durability, and limited biological activity, advances in gene-editing, including non-viral piggyBac transposon systems (e.g., P-PSMA-101, NCT04249947), enable enrichment of stem cell memory T cells (TSCM), enhancing CAR-T persistence while integrating safety switches (iCasp9) to mitigate fatal toxicities. Combinatorial approaches, including bi-specific CAR-T targeting dual antigens (PSMA + GD2, PSMA + CD70), PD-1 silencing, and co-stimulatory optimization (CD28, TNFRSF9/4-1BB), aim to improve tumor recognition, reduce immune escape, and enhance expansion within the suppressive tumor microenvironment. Integration of microenvironment-resistant designs (e.g., TGFβRDN, PD1-CD28 switch receptors) has shown early promise in enhancing persistence and efficacy, though further optimization is needed to prevent severe CRS and MAS events.

Comparatively, BiTEs on the other hand are particularly susceptible to T-cell exhaustion, anti-drug antibody formation, while ADC efficacy is limited by heterogeneous antigen expression and payload-specific resistance mechanisms. Importantly, while trials are continuing, all these approaches have yet to achieve durability to result in long-term remissions seen with cellular therapies in hematologic malignancies.

Future of CAR-T, relative to bites and ADCs

Looking forward, future CAR-T trials aim to prioritize integrated strategies combining CAR-T engineering, immune checkpoint modulation, microenvironment resistance, prophylactic cytokine control. Adaptive dosing, early biomarker-based toxicity monitoring, and the incorporation of safety switches will be essential to maximize efficacy while minimizing risk. Additionally, integrative approaches targeting multiple antigens or integrating checkpoint inhibition or even BiTEs may overcome antigen escape and tumor heterogeneity, key contributors to prior failures. Collectively, while prostate CAR-T therapy remains in an early yet promising stage compared to BiTEs and ADCs, the lessons from terminated or failed trials, and considering comprehensive approaches among these technologies may provide crucial guidance for designing next-generation constructs that are both potent and safe [59].

Abbreviations

4-1BB (CD137, TNFRSF9)	Costimulatory tumor necrosis factor receptor superfamily member 9
ADC	Antibody–drug conjugate
ADT	Androgen deprivation therapy
AE	Adverse event
ARPIs	Androgen receptor pathway inhibitors
B7-H3 (CD276)	B7 homolog 3 immune checkpoint molecule
BiTEs	Bispecific T-cell engagers
BRCA1/2	Breast cancer gene 1 and 2
CAR	Chimeric antigen receptor
CAR-T	Chimeric antigen receptor T-cell
CAR-T-PSMA-TGFβRDN	Chimeric antigen receptor T-cell targeting prostate-specific membrane antigen with dominant-negative transforming growth factor-β receptor
carHLH	CAR-T-cell–associated hemophagocytic lymphohistiocytosis
CD2	Cluster of differentiation 2
CD44v6	Cluster of differentiation 44 variant 6
CD70	Cluster of differentiation 70
CLDN6	Claudin-6
CRPC	Castration-resistant prostate cancer
CRS	Cytokine release syndrome
CXCL9	Chemokine (C-X-C motif) ligand 9
DLTs	Dose-limiting toxicities
dnTGFBR2	Dominant-negative transforming growth factor-β receptor 2
EpCAM (CD326)	Epithelial cell adhesion molecule
FDA	Food and Drug Administration
Flu/Cy	Fludarabine and cyclophosphamide
GD2	Disialoganglioside GD2
HRR	Homologous recombination repair
ICANS	Immune effector cell-associated neurotoxicity syndrome
iCasp9	Inducible caspase-9
IL-1	Interleukin-1
IL-1β	Interleukin-1 beta
IL-2	Interleukin-2
IL-2Rα	Soluble interleukin-2 receptor alpha
IL-4	Interleukin-4
IL-6	Interleukin-6
IL-8	Interleukin-8
iMC	Inducible MyD88/CD40 costimulatory domain
IP-10	Interferon-γ–induced protein 10
KLK2	kallikrein-2
MAPK/ERK	Mitogen-activated protein kinase/extracellular signal-regulated kinase
MAS	Macrophage activation syndrome
mCRPC	Metastatic castration-resistant prostate cancer
MeSH	Medical Subject Headings
MHC	Major histocompatibility complex
NK	Natural killer
NKG2D	Natural killer group 2 member D
NKG2DL	NKG2D ligands
NCCN	National Comprehensive Cancer Network
ORR	Overall response rate
PARP	Poly(ADP-ribose) polymerase
PCa	Prostate cancer
PD-1	Programmed cell death protein 1
PD-1–CD28	Programmed cell death protein 1–cluster of differentiation 28 switch receptor
PD-L1	Programmed cell death protein ligand 1
PI3K/Akt	Phosphoinositide 3-kinase/protein kinase B
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PSA	Prostate-specific antigen
PSCA	Prostate stem cell antigen
PSMA	Prostate-specific membrane antigen
RNA-LPX	RNA-lipoplex
STEAP2	Six-transmembrane epithelial antigen of the prostate 2
TGF-β	Transforming growth factor-β
TGFβDN	Dominant-negative transforming growth factor-β receptor
TIL	Tumor-infiltrating lymphocyte
TM	Target module
TNF	Tumor necrosis factor
TNFRSF9	Tumor necrosis factor receptor superfamily member 9
TNFSF14	Tumor necrosis factor superfamily member 14
TSCM	Stem cell memory T cells

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

F. H., N.S., V.B. were responsible for drafting of the text, sourcing, analyzing, investigation results, drawing original diagrams and algorithms, and critical revision for important intellectual content. F. H., N.S., V.B. gave final approval of the manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available in the"ClinicalTrials.gov, "Cochrane", and "PubMed" repository. We used the following MeSH keywords and their combinations: Advanced PubMed term: "Chimeric antigen receptor T-cell therapy" OR "Chimeric antigen receptor therapy" OR "Chimeric antigen receptor" OR "CAR" OR "CAR T-cell therapy" OR "CAR-T" OR "CAR-T Cancer Therapy" OR "CAR-T cancer treatment" OR "CAR-T therapy". The data was last extracted up to December 1, 2024. All clinical trials with outcomes that involved Prostate cancer with CAR-T cell therapy treatment were included in the data extraction. Initial data search with keywords mentioned above retrieved a total of 27,480 from PubMed records, 3,149 from Cochrane, and 1,936 from Clinicaltrials.gov.

Declarations

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Consent for publication

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Competing interests

The authors declare no competing interests.

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